Sahayak · सहायक · Ideas for India 2026

An offline scam shield and benefits guide for people new to digital money, in Hindi and English, by voice or touch.

"Is this a scam?" Paste, speak, photograph or describe a message, a call or a UPI QR code. In milliseconds: Scam, Suspicious, No scam signs (never "safe"), or Could not check (a language it cannot read yet: safe-default advice, no false green). Then the reasons, what to do (money lost: call 1930; an attempt: report on Chakshu) and a complaint draft. A QR card says: scanning sends money, never brings it in.

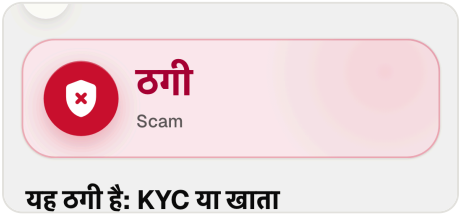
"What am I owed?" A short spoken interview finds which of 12 central schemes a person can get, health first: Ayushman Bharat pays up to ₹5 lakh a year for hospital admission (not OPD); at 70+, the Vay Vandana card (cover shared with a 70+ spouse). Then pensions, PMSBY, PMJJBY, APY, PM-KISAN, e-Shram, PM-SYM, Jan Dhan: the papers, the office, what to say, and a printed slip with a QR for the CSC operator.

HOW IT WORKS

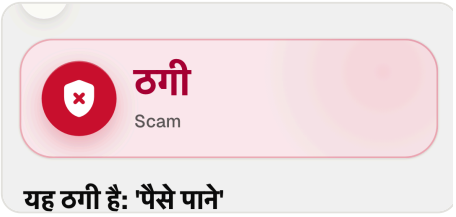
- **One node at a CSC or bank-agent counter** (a laptop or mini-PC). Phones join its Wi-Fi, which has no internet; the app is a cached web page, no install.
- **At home, too:** a phone that has opened Sahayak's secure web address once (the public site today; the node's own after a one-time HTTPS setup) keeps its own scam check and benefits interview, with no node or internet.
- **Named signals decide, not a model:** 64 signals in a signed content pack (lookalike links, .bank.in rule, UPI "scan to receive", OTP asks, digital arrest, "pay for your Ayushman card"…), a pattern matcher and a small classifier. Vetted templates explain each verdict; an optional local LLM (off by default) may add a line only through a safety gate.
- **Scheme rules as data**, each traced to its official page and date; yes / likely / unknown / no; asks only what can change a result.
- **Offline voice** both ways on the node; amounts spoken as words; "I heard …, is that right?" before any answer is used.
- **Operator console:** "Ask the agent" queue, assisted checks read aloud, case log, 58 mm slips, counters with no personal data.

MEASURED, NOT CLAIMED

Real messages	137 that people received, as published by the Income Tax portal, PIB Fact Check, courts and fact-checkers, scored once. Hindi, English, Hinglish: 67 / 97 scams caught (69%), 11 / 34 genuine flagged; keyword blocklist 62 and 17. Lower than on our own messages: the honest number
Scams caught	34 / 37 (91.9%; 95% Wilson CI 79%–97%) on a frozen test split of 60 messages our team wrote, vs 51.4% for a keyword blocklist; false alarms 2 / 23 (8.7%) vs 43.5%; 2 ms a verdict on a laptop CPU
Blind red team	182 messages by a separate AI model that never saw the code, scored once. Hindi, English, Hinglish: 92 / 103 scams caught, 14 / 59 hard genuine flagged (blocklist 51 and 25). Other languages: 5 / 15 caught, 10 "could not check", 0 false greens
An AI model instead?	qwen2.5:3b, zero-shot on the same 137: 99 / 103 scams caught but 21 / 34 genuine flagged (62%), 2.13 s a message on a GPU. Sahayak: 67 / 103 and 11 / 34 (32%), 1.7 ms
Scheme rules	2,400 / 2,400 decisions match a re-derivation by the same author (a second person's check is to come); median 5 questions, never more than 7
Hindi speech	15.4% word error on Google FLEURS Hindi (418 recordings of adults reading sentences aloud), offline; reply audio 0.65 s after the mic is released
Screenshots	same scam decision as the typed message on 51 / 55 rendered SMS screenshots, read offline
Under load	30 phones doing a whole visit at once, 300 checks at once and 5,000 in a row: 0 errors; 360 hostile inputs, 0 unexpected answers (Apple M3)
Offline	0 outbound connections or outside DNS lookups, counted live by an audit hook; a firewall blocks the rest
On the phone	the node's exact answer on 1,841 scam checks and 7,983 benefits cases (0 mismatches)







PRIVACY AND SAFETY

- Messages, voices and answers stay in the node's memory for at most 15 minutes; nothing typed or said is written to disk.
- Case log only with consent: no names or numbers, encrypted, deleted after 30 days, one "delete everything" button.
- Green means "no signs found, still verify", never "safe"; a language it cannot read gets a grey "could not check". Every amount shown comes from a signed pack. Content updates are Ed25519-signed; an altered pack is refused.

Team: Roshan Raj, Ishmiit Singh · **Try it:** ishmiit.github.io/Sahayak · **Dates:** prototype due 8 Oct 2026; Grand Jury Round, Delhi, 14 Oct · **Caveats:** ScamBench v0 and our red team were written by our team (optimistic); the blind set was written for the test, not received by real people; v1 adds real messages with consent; no field test with real users yet. Method and limits: docs/TESTING_REPORT.md. Hindi voices and the optional language model are licensed for non-commercial use.